

A Matter Of Speed

Solid state drives are no longer in the realm of only high end system builders. If a speedy system is what you are aiming for, an SSD is one component you should invest in. We are here as always to help out with the best of the best in the market



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One are the days when you could press the power button of your machine and go and listen to a song while waiting for your system to start. Hard drives are great and they will live long thanks to their economical pricing but there isn't a better time than now to start thinking about buying an SSD.

A solid state drive not only gives you faster speeds and faster data access times vis-a-vis spinning hard drives, it also helps in faster system startup and application loading. What's more, the cost per GB has come down quite a bit –

scaling the ₹200 mark in 2011, coming down to the ₹100 mark last year and finally this year seeing some drives touching costs per GB as low as ₹50. Thanks to the shrinking of the NAND chip-making process, prices have dipped dramatically.

Last year's SSD test had all the drives sporting the same SandForce controller, with difference being just in the type of NAND memory being used. This year we have a variety in the types of SSD controllers being employed by different SSD makers. Thanks to this, there is quite a variance in the performance numbers. Since

the last comparison test had all the SSDs sporting a SandForce SF2281 controller, there was no need for a DRAM for caching purpose. SandForce controller supports on-the-fly compression of data using the DuraWrite technology thereby limiting the number of times the flash memory is written to – in essence increasing the SSD's lifetime. However, this time around we had quite a lot of SSDs without the SandForce controller. Since the controllers on these SSDs do not support compression feature or have on-chip buffer, most of them had a cache buffer (either a DDR2 or DDR3 or SDRAM) to cache data.

FEATURES, DESIGN

SanDisk is quite a known name as far as NAND memory chips go, so it was only a matter of time before we saw an SSD from the NAND maker. The SanDisk Ultra Plus 256GB drive has a 7mm profile meant for use in notebooks. There is a spacer which can be stuck onto the SSD to provide a snug fit within the notebook. You also have a desktop version of the same SSD which basically bundles in a 3.5-inch tray. It comes in an all-black body with a simple sticker design on the front face.

On opening the SSD enclosure we found a quarter-profile PCB housing the SSD components. It sports a Marvell 88SS9175, a four-channel controller. Thanks to the four-